3-12-26 (WEEK 3)

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EXAMPLE

You open a xiaolongbao restaurant, and have a promotion to attract customers: if you toss a coin, if eight heads in a row, they can eat for free.

Then, $\mathbb{P}(\text{customer eats for free}) = \frac{1}{2^8}$, but if $\exists$ infinitely many customers, infinitely many get free food

EXAMPLE

i^{th} customer gets free food if they get i heads in a row

Say $E_i := \{i^{th} \text{ customer gets free food}\}$

Then, $\mathbb{P}(E_i) = \frac{1}{2^i} \Rightarrow \sum_{i=1}^{\infty} \mathbb{P}(E_i) = 1 < \infty$

$\therefore$ By the 1st Borel-Cantelli Lemma, $\mathbb{P}(\limsup_{i\to\infty} E_i) = 0$ (i.e. almost certainly, only finitely many customers get free food)

EXAMPLE

A fair coin which we toss n times: Let $H_n := \#$ heads

Let $E_n = \{H_n > (\frac{1}{2}+\varepsilon)n\}$ for some $\varepsilon > 0$

Consider $\mathbb{P}(E_n)$: $S = \{H, T\}^n$ ← outcomes of n coin tosses, Head or Tail

Then, $E_n \subseteq S = \{w \in \{H,T\}^n \mid w \text{ has } \geq (\frac{1}{2}+\varepsilon)n \text{ Heads}\}$

$$= \bigsqcup_{r=(\frac{1}{2}+\varepsilon)n}^{n} F_r, \text{ where } F_r = \{w \mid w \text{ has exactly } r \text{ heads}\}$$

$$\Rightarrow \mathbb{P}(E_n) = \sum_{r=(\frac{1}{2}+\varepsilon)n}^{n} \mathbb{P}(F_r)$$

We know $\mathbb{P}(F_r) = \frac{|F_r|}{|S|}$

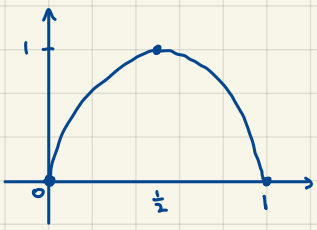
By product rule, $|S| = 2^n$

$|F_r| = |\{(w_1, \dots, w_n) \mid \text{exactly } r \text{ of the } w_i = H\}| = \binom{n}{r}$ ← choose r out of n to be the set of heads

$$\therefore \mathbb{P}(F_r) = \frac{\binom{n}{r}}{2^n} \Rightarrow \mathbb{P}(E_n) = \sum_{r=(\frac{1}{2}+\varepsilon)n}^{n} \frac{\binom{n}{r}}{2^n} = 2^{-n} \sum_{r=(\frac{1}{2}+\varepsilon)n}^{n} \binom{n}{r}$$

DEFINITION

The binary entropy function is defined as $H(x) = -x\log_2 x - (1-x)\log_2(1-x)$ for $0 \leq x \leq 1$



FACT

① $\sum_{r=(\frac{1}{2}+\varepsilon)n}^{n} \binom{n}{r} \approx 2^{H(\frac{1}{2}+\varepsilon)n}$ for $\varepsilon \geq 0$

② $H(x) \leq 1$, with "=" iff $x = \frac{1}{2}$

BACK TO EXAMPLE

Then, if $H(\frac{1}{2}+\varepsilon) = 1-\delta$, $\delta > 0$, thus $\mathbb{P}(E_n) \leq 2^{-n} \cdot 2^{(1-\delta)n} = 2^{-\delta n}$

$\therefore \sum_{n=1}^{\infty} \mathbb{P}(E_n) \leq \sum_{n=1}^{\infty} 2^{-\delta n} < \infty$ since $2^{-\delta} < 1$

By the 1st Borel-Cantelli Lemma, $\mathbb{P}(\limsup_{n\to\infty} E_n) = 0 \Leftrightarrow \mathbb{P}(E_n : \text{infinitely often}) = 0 \Leftrightarrow \mathbb{P}(\exists N, \text{ s.t. } \forall n \geq N, \# \text{ heads in first } n \text{ tosses} < (\frac{1}{2}+\varepsilon)n) = 1$

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Let $h_n := \frac{\#\text{heads in } n \text{ tosses}}{n}$

We showed $\forall \varepsilon > 0$, with prob 1, $\exists N$ s.t. if $n > N$, then $h_n \leq \frac{1}{2} + \varepsilon$

$\Rightarrow \limsup_{n\to\infty} h_n \leq \frac{1}{2}$

However, by symmetry, for $t_n = \frac{\#\text{tails in } n \text{ tosses}}{n}$, we have $\limsup_{n\to\infty} t_n \leq \frac{1}{2}$

$\because h_n + t_n = 1 \Leftrightarrow \liminf_{n\to\infty} h_n \geq \frac{1}{2}$

$\therefore \lim_{n\to\infty} h_n = \frac{1}{2}$

CONDITIONAL PROBABILITIES AND INDEPENDENCE

DEFINITION

Let $(S, \mathbb{P})$ be a probability space and let $F \subseteq S$ be an event s.t. $\mathbb{P}(F) > 0$

The conditional probability of an event E given F is $\mathbb{P}(E|F) := \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}$

EXAMPLE

We roll a fair die.

Let $E = \{\text{outcome is even}\}$, then $\mathbb{P}(E) = \frac{1}{2}$

For $F = \{\text{outcome is prime}\}$, then $\mathbb{P}(F) = \frac{1}{2}$

Suppose we know the outcome is prime. What is the probability it is even?

Then, $\mathbb{P}(E|F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)} = \frac{\mathbb{P}(\{2\})}{\mathbb{P}(F)} = \frac{1/6}{1/2} = \frac{1}{3}$

PROPERTIES

(1) $0 \leq \mathbb{P}(E|F) \leq 1 \ \forall E \subseteq S$ ($\because 0 \leq \mathbb{P}(E \cap F) \leq \mathbb{P}(F)$, as $E \cap F \subseteq F$)

(2) $\mathbb{P}(F|F) = \frac{\mathbb{P}(F \cap F)}{\mathbb{P}(F)} = \frac{\mathbb{P}(F)}{\mathbb{P}(F)} = 1$. In fact, $E \supseteq F \Rightarrow \mathbb{P}(E|F) = 1$. In particular, $\mathbb{P}(S|F) = 1$

(3) If E and F are disjoint (i.e. $E \cap F = \emptyset$), $\mathbb{P}(E|F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)} = \frac{\mathbb{P}(\emptyset)}{\mathbb{P}(F)} = 0$. In particular, $\mathbb{P}(\emptyset|F) = 0$

(4) $\mathbb{P}(E \cap F) = \mathbb{P}(E|F)\,\mathbb{P}(F)$ by def

(5) If $E_1, E_2, E_3, \ldots$ are mutually exclusive, then $\mathbb{P}(\bigcup_{i=1}^{\infty} E_i | F) = \sum_{i=1}^{\infty} \mathbb{P}(E_i|F)$ since $\mathbb{P}(\bigcup_{i=1}^{\infty} E_i | F) = \frac{\mathbb{P}((\bigcup_{i=1}^{\infty} E_i) \cap F)}{\mathbb{P}(F)} = \frac{\mathbb{P}(\bigcup_{i=1}^{\infty} (E_i \cap F))}{\mathbb{P}(F)} = \frac{\sum_{i=1}^{\infty} \mathbb{P}(E_i \cap F)}{\mathbb{P}(F)} = \sum_{i=1}^{\infty} \frac{\mathbb{P}(E_i \cap F)}{\mathbb{P}(F)} = \sum_{i=1}^{\infty} \mathbb{P}(E_i|F)$

still mutex

countable additivity

COROLLARY

$(S, \mathbb{P}(\cdot|F))$ is also a probability space, since

① $0 \leq \mathbb{P}(E|F) \leq 1 \ \forall E \subseteq S$

② $\mathbb{P}(S|F) = 1$

③ Countable additivity

EXAMPLE

n people at a party, each with their own hat. At the end of the party, everyone gets a random hat.

What is the probability that exactly r people get their own hat?

$S = S_n = \{\text{permutations of } n\}$

$\mathbb{P}$ = uniform probability measure

Let $E_r = \{\text{exactly } r \text{ people get their own hats}\}$

Notice,

$E_r = \bigsqcup_{\substack{R \subseteq [n] \\ |R| = r}} E_R$ where $E_R = \{\text{people in } R \Leftrightarrow \text{people get their own hats}\}$

$\Rightarrow \mathbb{P}(E_r) = \sum_{\substack{R \subseteq [n] \\ |R| = r}} \mathbb{P}(E_R)$

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By symmetry, E_R has the same probablity $\forall R$

$\Rightarrow \mathbb{P}(E_r) = \binom{n}{r}\mathbb{P}(E_{\{n-r+1, n-r+2, \dots, n\}})$

Notice, $E_{\{n-r+1,\dots,n\}} = \underbrace{\{\text{people } n-r+1, \dots, n \text{ get their own hat}\}}_{E} \cap \underbrace{\{\text{people } 1, \dots, n-r \text{ don't get their own hat}\}}_{F}$

We know:

$$\mathbb{P}(F) = \frac{|F|}{|S_n|} = \frac{(n-r)!}{n!}$$

permute hats $1, \dots, n-r$; fix hats $n-r+1, \dots, n$

As $\mathbb{P}(E|F) = \mathbb{P}(\text{hats } 1, \dots, n-r \text{ are shuffled w/ nobody getting their own hats}) = \mathbb{P}(\text{derangement of } n-r \text{ hats})$

$\mathbb{P}(E|F)\mathbb{P}(F) = \mathbb{P}(E \cap F)$

By prev results, $\mathbb{P}(E|F) = \sum_{k=0}^{n-r} \frac{(-1)^k}{k!} \Rightarrow \mathbb{P}(E_{\{n-r+1,\dots,n\}}) = \frac{(n-r)!}{n!}\sum_{k=0}^{n-r}\frac{(-1)^k}{k!}$

$\therefore \mathbb{P}(E_r) = \binom{n}{r}\mathbb{P}(E_{\{n-r+1,\dots,n\}}) = \binom{n}{r}\frac{(n-r)!}{n!}\sum_{k=0}^{n-r}\frac{(-1)^k}{k!} = \frac{1}{r!}\sum_{k=0}^{n-r}\frac{(-1)^k}{k!} \xrightarrow{n-r\to\infty} \frac{1}{r!}\cdot\frac{1}{e}$

THE LAW OF TOTAL PROBABILITY

DEFINITION

Given a set S, we say $E_1, \dots, E_n$ partition S if $E_1 \sqcup E_2 \sqcup \dots \sqcup E_n = S$

THEOREM (LAW OF TOTAL PROBABILITY)

Let $(S, \mathbb{P})$ be a probability space and let $S = F_1 \sqcup F_2 \sqcup \dots \sqcup F_n$ be a partition

Then, for any event E, $\mathbb{P}(E) = \sum_{i=1}^{n}\mathbb{P}(E|F_i)\mathbb{P}(F_i)$

<u>Proof</u>

As $E = E \cap S$, thus $E = E \cap (\bigsqcup_{i=1}^{n} F_i) = \bigsqcup_{i=1}^{n}(E \cap F_i)$

$\therefore \mathbb{P}(E) = \sum_{i=1}^{n}\mathbb{P}(E \cap F_i) = \sum_{i=1}^{n}\mathbb{P}(E|F_i)\mathbb{P}(F_i)$ $\square$